

**Consensus fusion weights — per-modality contribution to the final fused model**  
Mean  $\pm$  SD of the elastic-net fusion coefficient across outer CV folds

fusion coefficient (mean  $\pm$  SD across folds)

**Global**

**DHP**

**T-DM1**

fusion coefficient (mean  $\pm$  SD across folds)

fusion coefficient (mean  $\pm$  SD across folds)

| Modality | Mean  | SD | Selection Rate |
|----------|-------|----|----------------|
| Clin     | +0.27 |    | sel 46%        |
| RNA      | +2.50 |    | sel 100%       |
| DNA      | +0.37 |    | sel 38%        |
| Prot     | +0.69 |    | sel 71%        |
| WSI      | +1.04 |    | sel 52%        |

| Modality | Mean  | SD | Selection Rate |
|----------|-------|----|----------------|
| Clin     | +0.63 |    | sel 38%        |
| RNA      | +1.26 |    | sel 92%        |
| DNA      | +0.54 |    | sel 75%        |
| Prot     | +1.87 |    | sel 97%        |
| WSI      | +0.91 |    | sel 48%        |

| Modality | Mean  | SD | Selection Rate |
|----------|-------|----|----------------|
| Clin     | +1.13 |    | sel 77%        |
| RNA      | +1.67 |    | sel 91%        |
| DNA      | +1.83 |    | sel 84%        |
| Prot     | +1.39 |    | sel 70%        |
| WSI      | +0.26 |    | sel 26%        |